

Naveen Gurupatham J

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Education

Indian Institute of Technology Delhi

Ph.D.

Dec 2021 - present

New Delhi, India

Mepco Schlenk Engineering College

B.Tech

Jul 2016- Apr 2020

Sivakasi, Tamilnadu, India

Experience

Indian Institute of Technology Delhi

Junior Research Fellow

Oct 2020 - Nov 2021

New Delhi, India

About me

I worked on diverse research projects, including multi-variate model development, large language models development, process modeling, model predictive control, digital twin development, cyber-physical system development and implementation, and Process optimization.

Publications and Patent

- **NG Jesubalan**, G Thakur and AS Rathore "Deep neural networks for prediction and control of permeability decline in single pass tangential flow ultrafiltration in continuous processing of monoclonal antibodies", *Frontiers in Chemical Engineering* 5 (2023).
- **NG Jesubalan**, N Saxena, V B Yezhuvath, N Deore and AS Rathore "AI-enhanced continued process verification for Ultrafiltration/Diafiltration", *Biotechnology and Bioengineering* (2025).
- **NG Jesubalan**, N Saxena, S Banerjee, N Deore, K Gupta and AS Rathore "Q-learning agent-based level control of surge tanks for improved continuous manufacturing of monoclonal antibodies", *Computers and Chemical Engineering* (2025).
- **NG Jesubalan**, R Sharma, AS Rathore, "Application of ensemble learning to augment fluorescence-based PAT and enable real-time monitoring of protein refolding", *Biochemical Engineering Journal* (2024)
- Anupa Anupa, **NG Jesubalan**, R Trivedi, Nitika Nitka, VS Buddhiraju, V Runkana, AS Rathore, "Implementation of machine learning tool for continued process verification of process chromatography unit operation", *Journal of Chromatography A* (2025)
- **NG Jesubalan**, N Saxena, VB Yezhuvath, N Deore, AS Rathore, "AI-enhanced continued process verification for Ultrafiltration/Diafiltration", *Biotechnology and Bioengineering* (under review)
- S Nikita, G Thakur, **NG Jesubalan**, A Kulkarni, VB Yezhuvath and AS Rathore, "AI-ML applications in bioprocessing: ML as an enabler of real time quality prediction in continuous manufacturing of mAbs," *Computers and Chemical Engineering* 164 (107896) (2022).
- AS Rathore, **NG Jesubalan**, and G Thakur, "Perspectives on Process Analytical Technology," *BioPharm International* 35 (5) (2022).
- H Chhabra, **NG Jesubalan** and AS Rathore, "Rapid detection of trace chlorine dioxide (ClO₂) concentration in liquid sample using a soft sensor", *Water Research* 242 (120231) (2023)
- AS Rathore, S Nikita and **NG Jesubalan**, "The role of Soft Sensors in Monitoring and Control of Downstream Processing for Production of Biotherapeutic Products," *Biosensors and Bioelectronics* X 12 (100263) (2022)
- S Banerjee, **NG Jesubalan**, A Kulkarni, A Agarwal, AS Rathore, "Developing cyber-physical system and digital twin for smart manufacturing: Methodology and case study of continuous clarification," *Journal of Industrial Information Integration* (38) (100577) (2024)
- S Banerjee, S Nikita, **NG Jesubalan**, A Kulkarni, K Gupta, AS Rathore, "Holistic process control framework for smart biomanufacturing: A deep learning driven approach," *Computers and Chemical Engineering* (3187) (108742) (2024)
- S Banerjee, **NG Jesubalan**, S Nikita, AS Rathore, "Multi-objective Optimization of Continuous CHO Cell Clarification using Acoustic Wave Separation", *Separation and Purification Technology* (2024)
- S Banerjee, S Mandal, **NG Jesubalan**, R Jain, AS Rathore, "NIR spectroscopy—CNN-enabled chemometrics for multianalyte monitoring in microbial fermentation", *Biotechnology and Bioengineering* (121) (1803-1819) (2024)
- **NG Jesubalan**, R Sharma, AS Rathore, "Multi-objective optimization (MOO) of insulin refolding reaction using ensemble coupled NSGA-II algorithm", *Separation and Purification* (Under revision)
- Patent: **NG Jesubalan**, H Chhabra, "Rapid detection on trace components using NIR and FTIR based PAT tools"

Awards and Certifications

- Received the Innovation Pride award from TCS for contribution to the project “**Digitalization of Biopharmaceutical Manufacturing.**”
- Awarded **third prize** in an oral presentation entitled “Uses analytical tools for the characterization of biotherapeutics” at the International Conference on Clinical Research, 2020, organized by the School of Biosciences and Technology, Vellore Institute of Technology, Vellore, Tamil Nadu.
- Completed Coursera project network on **Deep Learning with PyTorch: Image Segmentation**
- Successfully cleared competitive exam - **GATE Biotechnology 2021.**
- Language proficiency exam - **BEC Vantage level.**
- Completed Coursera certification on **Supervised Machine Learning: Regression and Classification**
- Completed Coursera certification on **Advanced Learning Algorithms**
- Completed Coursera certification on **Unsupervised Learning, Recommenders, Reinforcement Learning**
- Completed Coursera certification on **Neural Networks and Deep Learning**
- Completed a National Programme on Technology Enhanced Learning (NPTEL) course on **Biomedical Nanotechnology, Principles of Downstream processing of Biotechnology and Functional Genomics**
- Completed an NPTEL course on **Python for Data sciences** by Prof. Raghunathan Rengasamy, Indian Institute of Technology, Madras, and obtained Elite certificate.

Technical Exposure

- Presented my research work in **ACS Spring 2026 and ACS Spring 2025** in both oral and poster format.
- Presented my research work in **BPI 2024** in oral format.
- Presented my research work in **ChemRD2**, an institute level event and secured third position in poster presentation.
- Presented my research work in **SIRe Symposium 2025** in both oral format.
- Presented my research work in **SIRe Symposium 2026** in both poster format.
- Participated in **Merck innovation Cup 2024: Team smart manufacturing.** During this period, our team presented a cutting-edge tool aimed at reducing the divide between R and D and manufacturing in chemical production, thereby laying the groundwork for Industry 4.0.
- Delivered a Technical Talk on the Role of ML in continuous manufacturing of Biotherapeutic manufacturing and a two-day Hands-on Course in Explainable AI and Advanced Machine Learning in Bioprocessing as part of the Annual **CBT Course Series**, held from December 11th to 15th, 2023.
- Delivered a Technical Talk on the Role of DCS in continuous manufacturing of Biotherapeutic manufacturing as part of the Annual **CBT Course Series**, held from December 11th to 15th, 2023.
- Delivered a Technical Talk on the Role of PAT tools in continuous manufacturing of Biotherapeutic manufacturing as part of the Annual **CBT Course Series**, held from December 2024.
- Served as a Secretary for the **Science Forum Club** in our institute and organized various events in **Gyan Mitra 2019** (inter-college technical symposium) and **RADIX'19** (State level symposium for students)
- Serving as a Webmaster for **ACS-Biot division India**
- Demonstrated a model titled Visualization of DNA using gel electrophoresis during the **mepexpo'19** on August 2 and 3, 2019

Hobbies

- Playing Piano
- Running
- Photography